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Bihar Engineering University, Patna

B.Tech 3rd Semester Exam., 2020 (New Course)

Code: 100311 ***Subject: MATHEMATICS-III (Differential Calculus)*** ***Time: 03 Hours*** ***Full Marks: 70***

Instructions:-

(i) *The marks are indicated in the right-hand margin.*
(ii) *There are NINE questions in this paper.*
(iii) *Attempt FIVE questions in all.*
(iv) *Question No. 1 is compulsory.*

Q.1 Choose the correct answer of the following (any seven): **[14]**

(a) The value of $\lim_{x\to 0}(\sin x/x)^{1/x}$ is

(i) 0 (iii) e
(ii) 1 (iv) $1/e$

(b) Let $f(x)=|x|$ and $g(x)=|x^3|$, then

(i) $f(x)$ and $g(x)$ both are continuous at $x=0$
(ii) $f(x)$ and $g(x)$ both are differentiable at $x=0$
(iii) $f(x)$ is differentiable but $g(x)$ is not differentiable at $x=0$
(iv) $f(x)$ and $g(x)$ both are not differentiable at $x=0$

(c) The value of $\nabla^2[(1-x)(1-2x)]$ is equal to

(i) 2 (iii) 4
(ii) 3 (iv) 6

(d) If $v= xy^2i- 2x^2yzj- 3yz^2k$, then the value of curl v at (1,1,1) is equal to

(i) $-(j-2k)$ (iii) $-(i-2k)$
(ii) $-(i-3k)$ (iv) $(i-2j-k)$

(e) The degree of the differential equation $y(dx/dy)+(dx/dy)^2+\sin y(dx/dy)^3-\cos x = 0$ is

(i) 0 (iii) 2
(ii) 1 (iv) Cannot be determined

(f) The solution of the boundary value problem $(x-y^2x)dx - (x^2y-y)dy = 0$, $y(0)=0$ is

(i) $x^2-y^2=0$ (iii) $x-2y=0$
(ii) $2x-y=0$ (iv) None of the above

(g) Let $P_n(x)$ be the Legendre polynomial of degree $n\geq 0$. If $\int_{-1}^{1}P_{n-1}^{2}(x)dx =[2] / [4n-1]$ then the value of (k, l) is

(i) (1, 1) (iii) (2, 1)
(ii) (1, 2) (iv) (2, 2)

(h) The general solution of Bessel differential equation $x^2y''(x) + xy'(x) + (x^2-64)y(x) = 0$ is

(i) $y = AJ_8(x) + BJ_{-8}(x)$, where A and B are arbitrary constants
(ii) $y = AJ_8(x) + BY_{-8}(x)$, where A and B are arbitrary constants
(iii) $y = AJ_8(x) + J_{-8}(x)$, where A is arbitrary constant
(iv) $y = J_{3/4}(x) + Y_{3/4}(x)$

(i) The equation $p\tan y + q\tan x =\sec^2 z$ is of order

(i) 1 (iii) 0
(ii) 2 (iv) None of the above

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(j) The solution of $p\tan x + q\tan y = \tan z$ is

(i) $\sin x / \sin y = \varphi(\sin y / \sin z)$

(ii) $\sin x \cdot \sin y = \varphi(\sin y / \sin z)$

(iii) $\sin x / \sin y = \varphi(\sin y, \sin z)$

(iv) $\sin x / \sin y = \varphi(\sin y \cdot \sin z)$

Q.2 Solve both questions : **[14]**

(a) If $y = (\sin^{-1}x)^2$, then show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$. Hence find $(y_n)_0$.

(b) Find the value of $\lim_{x \to 0}(\tan x/x)^{1/x^2}$.

Q.3 Solve both questions : **[14]**

(a) Discuss the continuity of the following function $f(x,y)$ at point (0, 0):
$f(x,y) = [\sin\sqrt{(|xy|)} - \sqrt{(|xy|)}] / [\sqrt{(x^2+y^2)}]$, & $(x,y) \neq (0,0)$
0, & $(x,y) = (0,0)$

(b) For the function $f(x,y) = [xy(2x^2-3y^2)] / [x^2+y^2]$, & $(x,y) \neq (0,0)$
0, & $(x,y) = (0,0)$ check whether $f_{xy}(0,0)$ and $f_{yx}(0,0)$ are equal or not.

Q.4 Solve both questions : **[14]**

(a) Find the minimum value of $x^2+y^2+z^2$ subject to the condition $xyz=a^3$.

(b) Obtain the second-order Taylor's series approximation to the function $f(x,y) = xy^2 + y\cos(x-y)$ about the point (1, 1).

Q.5 Solve both questions : **[14]**

(a) If $f = (x^2+y^2+z^2)^{-n}$, then find div grad f and determine n, if div grad $f = 0$.

(b) Verify Green's theorem for $\int_C[(xy+y^2)dx + x^2dy]$ where C is bounded by $y=x$, $y=x^2$.

Q.6 Solve both questions : **[14]**

(a) Find the value of n for which the vector $r^n r$ solenoidal, where $r = xi + yj + zk$.

(b) Solve the differential equation $(y^4+2y)dx + (xy^3+2y^4-4x)dy = 0$.

Q.7 Solve both questions : **[14]**

(a) Solve $p = \sin(y-xp)$. Also find its singular solution.

(b) Solve $x^2d^2y/dx^2 - 3xdy/dx + 5y = x\log x$.

Q.8 Solve both questions : **[14]**

(a) Prove that $2nJ_n(x) = x(J_{n-1}(x) + J_{n+1}(x))$.

(b) Prove that $\sum_{n=0}\backslash infty[x^{n+1}] / [n+1]P_n(1) = 1/2\log([1+x] / [1-x])$.

Q.9 Solve both questions : **[14]**

(a) Solve $x^2p + y^2q = (x+y)z$.

(b) Solve $(x+y)(p+q)^2 + (x-y)(p-q)^2 = 1$.

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